Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

B.Sc. (Engg.) Fourth Year, First Semester Final Examination-2024

Course Code: EEE 475 Session: 2020-2021

Course Title: RF and Microwave Engineering

Total Marks: 100 **Credits: 3.0** **Total Time: 3 Hours**

*[Answer any **two** questions from each part. The figure in the right margin indicate full marks.]*

Part-A

1. (a) From the lumped-element equivalent circuit, derive the expression of the propagation constant in terms of series resistance R, series inductance L, shunt conductance G, and shunt capacitance C per unit length. Deduce this expression to low-loss transmission line. 8

   (b) If the propagation constant of a transmission line is $50\angle 89.38°$ m^{-1}, $R = 4\ \Omega$ m^{-1}, $L = 0.5\ \mu$H m^{-1}, $C = 200$ pF m^{-1}, calculate the value of shunt conductance with proper unit. 5

   (c) In Fig. 1, a lossless transmission line is shown where, $V(z) = V_0^+(e^{-j\beta z} + \Gamma e^{j\beta z})$, $I(z) = \frac{V_0^+}{Z_0}(e^{-j\beta z} - \Gamma e^{j\beta z})$, Γ is the reflection coefficient at point $z = 0$, Z_0 is the characteristics impdedance of the line, Z_L is the load impedance, and β is the phase constant. Show that

   (i) $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$

   (ii) $Z_{in}(l) = \left(\frac{Z_L - jZ_0 \tan\beta l}{Z_0 - jZ_L \tan\beta l}\right) Z_0$ 5+7

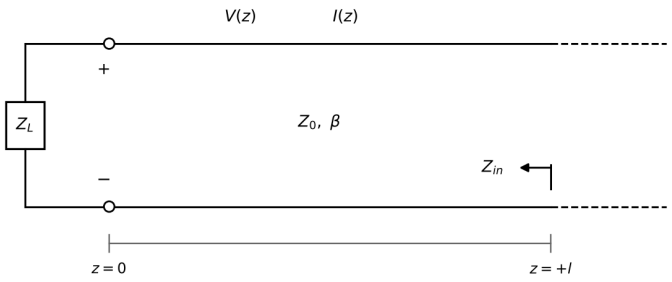


Fig. 1 of the paper (redrawn)

2. (a) Derive the equation of the constant resistance and constant reactance circle of a Smith Chart. You should start form the relation between the reflection coefficient and the normalized load. 12

   (b) A 75 Ω lossless transmission line is connected to a load impedance of $(35 - j50)\ \Omega$. Find the length position, l_p, and the length, l_s of a short-circuited stub required to match the load. You need to show two possible solutions using Smith Chart. 8

   (c) Explain how quarter-wave transformer can be used for impdeance matching. Mention its limitations. 5

3. (a) A plane wave in a nonmagnetic medium has $\vec{E} = 50\sin(10^8 t + 2z)\hat{a}_y$ V/m. Find:

   (i) The direction of wave propagation

   (ii) λ, f, and ϵ_r

   (iii) $\vec{H}$ 6

   (b) A telephone line has $R = 30\ \Omega$ km^{-1}, $L = 100$ mH km^{-1}, $G = 0$, and $C = 20\ \mu$F km^{-1}. At $f = 1$ kHz, obtain:

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(i) The characteristic impedance of the line
(ii) The propagation constant
(iii) The phase velocity 6

(c) An antenna with an impednace of $40 + j30\ \Omega$ is to be matched to a $100\ \Omega$ lossless line with a shorted stub. Determine:

(i) The required stub admittance
(ii) The distance between the stub and the antenna
(iii) The stub length
(iv) The standing wave ratio on each segment of the system 6

(d) A rectangular waveguide with dimensions $a = 2.5$ cm, $b = 1$ cm is to operate below 15.1 GHz. How many TE and TM modes can the waveguide transmit if the guide is filled with a medium characterized by $\sigma = 0$, $\epsilon = 4\epsilon_0$, $\mu_r = 1$? Calculate the cutoff frequencies of the modes. 7

Part-B

4. (a) The magnetic vector potential of a Hertzian dipole is given by: $\vec{A} = \frac{\mu_0 Idl}{4\pi r} e^{-j\beta r}\hat{z}$. Show that magnetic field, $H = \frac{Idl}{4\pi}\cos(\theta)\left[\frac{j\beta}{r} + \frac{1}{r^2}\right] e^{-j\beta r}\hat{\phi}$ [*sic*: should be $\sin(\theta)$]. 8

(b) Using the expression of H found in 4(a), show that the electric field component $E = \frac{\eta Idl}{4\pi}\cos(\theta)\left[\frac{j\beta}{r} + \frac{1}{r^2} - \frac{j}{\beta r^3}\right] e^{-j\beta r}\hat{\theta} + \frac{\eta Idl}{2\pi}\cos(\theta)\left[\frac{1}{r^2} - \frac{j}{\beta r^3}\right] e^{-j\beta r}\hat{r}$ [*sic*: the $\hat{\theta}$ term should have $\sin(\theta)$] 12

(c) Find the expression of average power density of Hertzian dipole in far field zone. 5

5. (a) An RF signal at $f = 100$ MHz is mixed with local oscillator signal at $f_{LO} = 90$ MHz.

(i) Calculate the IF (intermediate frequency) output.
(ii) Why are active filters used after mixing in a super-heterodyne receiver? 6

(b) For a 1 kHz message signal modulating a 1 MHz carrier with 80% modulation index in AM, calculate the total transmitted power if carrier power is 100 W. 5

(c) For an antenna with gain 10 dB transmitting 10 W power at 2.4 GHz, calculate the effective isotropic radiated power (EIRP) of the antenna. 5

(d) Write short notes (any three).

(i) Klystron
(ii) Magnetron
(iii) Gunn Diode Oscillator
(iv) Travelling Wave Tube (TWT) 3+3+3

6. (a) What are the uses of a microstrip line? Mention its advantages. Describe the type of losses in a microstrip line. 5

(b) Derive Jefeminko's equation and from there, explain the necessity of accelerating charge for electromagnetic radiation. 13

(c) A rectangular cavity resonator has dimensions of $a = 5$ cm, $b = 2$ cm, and $d = 15$ cm. Compute the resonant frequency of the dominant mode for a dielectric-field cavity of $\epsilon_r = 2.56$. 3

(d) What do you mean by effective height and effective area of an antenna? 4

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List of the relevant equations:

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

$$\nabla \times \mathbf{A} = \frac{1}{r^2\sin\theta} \begin{vmatrix} a_r & ra_\theta & (r\sin\theta)a_\phi \\ \dfrac{\partial}{\partial r} & \dfrac{\partial}{\partial\theta} & \dfrac{\partial}{\partial\phi} \\ A_r & rA_\theta & (r\sin\theta)A_\phi \end{vmatrix}$$

$$= \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial\theta}(A_\phi\sin\theta) - \frac{\partial A_\theta}{\partial\phi}\right]a_r + \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial A_r}{\partial\phi} - \frac{\partial}{\partial r}(rA_\phi)\right]a_\theta + \frac{1}{r}\left[\frac{\partial}{\partial r}(rA_\theta) - \frac{\partial A_r}{\partial\theta}\right]a_\phi$$

$$\begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} a_r \\ a_\theta \\ a_\phi \end{bmatrix}$$

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